GAME DESIGN DOCUMENT (GDD): DAHDI

Document Version: 1.0 (Production Master)
Project: Dahdi — Traditional Indian Mills Board Game
Institution: Academic / College Game Development Project
Date: September 2026

PART 1: CONCEPT DOCUMENT

Title Page

- Game Name: Dahdi: Traditional Indian Mills Board Game
- Game Logo: Dahdi title presented in the web interface; no separate logo asset is required.
- Game Catch Phrase: "Place wisely, form the mill, capture strategically."
- Document Type: Full Game Design Document (GDD) & Technical Specification
- Document Version: 1.0 (Gold Master Specification)

Credit Page

- Document Purpose: Authoritative game design and technical blueprint for the digital adaptation of the traditional Indian mills-style strategy game.
- Working Title: Project Dahdi Core
- Game Concept: Digital preservation and gamification of a traditional alignment-and-capture board game, implemented as a browser-based strategy experience.
- Author / Developer: Soumya Ranjan Samal
- Academic Project: College Game Development Project

Sign-Off

Role

Role	Name	Status	Signature Date
Student Developer	Soumya Ranjan Samal	Prepared	September 2026
Project Supervisor	Faculty / Project Supervisor	Pending	—
Project Evaluator	Faculty / Department	Pending	—

Introduction

Dahdi is a digital adaptation of a traditional Indian mills/alignment-style board game. Players place and move nine pieces on a 24-point board arranged as three concentric squares. The central tactical objective is to create lines of three pieces, called mills, and use those formations to capture eligible opposing pieces.

- Genre: Casual Digital Board Game / Traditional Strategy Game.
- Player Type: Casual players, students, families and strategy-game learners.
- Technical Form: Browser-based 2D interface using HTML5, CSS3 and JavaScript modules.
- Theme: Traditional Indian-inspired board-game presentation.
- Design Intentions: Preserve the strategic character of a mills game while making the rules understandable and convenient to play digitally.

Game Analysis

Game Description

Dahdi is a turn-based two-player strategy board game. Player 1 is represented as Peacock and Player 2 as Haldi. In Player vs AI mode, Haldi is controlled by the computer. The board contains 24 playable positions arranged across three concentric squares. Players alternate turns and interact directly with the board points while the rules engine maintains the authoritative match state.

Genre & Elements

- Genre: Traditional Board / Strategy / Alignment / Casual.
- Core Game Elements: 24 board points, nine pieces per side, mills, captures, movement, flying and win conditions.
- Game Content: Turn-based planning, tactical blocking, mill formation and piece reduction.
- Modes: Player vs AI and 2-Player Pass & Play.

Theme & Style

- Theme: Traditional Indian-inspired board-game heritage.
- Style: Clean 2D digital board presentation with clear tokens and legal-move indicators.
- Game Sequence: Placement → Movement → Flying, with capture decisions after mills.
- Player Count: Two sides; one can be AI-controlled.

Game Reference & Taxonomy

- Taxonomy: Deterministic turn-based strategy and alignment game.
- Strategic Immersion: Players evaluate immediate mills, threats, blocking positions and mobility.
- Learning Immersion: The rules are introduced through visible phases, indicators and the Rules screen.

Game Technical

- Technical Form: 2D top-down browser board with DOM/SVG/CSS based visual presentation.
- Camera View: Fixed full-board view so all 24 playable positions remain visible.
- Target Platform: Modern desktop and mobile web browsers.
- Technology: HTML5, CSS3 and JavaScript ES modules.
- Audio: Web Audio API.
- Input: Mouse/touch interaction and keyboard shortcuts.
- Storage: Browser localStorage for selected preferences.

Game Sales & Distribution

- Consumer Group: Students, casual strategy players, families and users interested in traditional games.
- Distribution: Browser-based academic demonstration and local web deployment.
- Monetization: Not part of the academic implementation; the project is designed as a free educational game.

Game Atmosphere

The atmosphere is intentionally calm and board-game focused. The interface combines a traditional-inspired board with modern controls, keeping attention on strategic decisions rather than fast action. Optional Tanpura ambience and soft action sounds provide an Indian-inspired acoustic identity.

Gameplay Walkthrough

1. Match Initialization

Start the game and choose Player vs AI or 2-Player Pass & Play. If AI mode is selected, choose Easy, Medium or Master. The optional diagonal board configuration can also be enabled from Settings.

2. Placement Phase

Each player has nine pieces. Players alternate placing one available piece on an empty board point. A newly completed line of three pieces forms a mill.

3. Capture

After a mill is formed, the player selects an eligible opponent piece for removal. Protected pieces inside mills are normally preserved while other removable pieces exist.

4. Movement Phase

After all pieces are placed, players move a selected piece along a legal connected board path to an empty point.

5. Flying Phase

When a player has exactly three pieces remaining, the Flying phase permits that player to move a piece to any empty point.

6. Winning Condition

The match ends when the opponent has been reduced to fewer than three pieces or has no legal movement available.

Key Features & Selling Features

- Dual Play Modes: Player vs AI and local Pass & Play.
- Three AI Levels: Easy, Medium and Master.
- Strategic Utilities: Undo, Hint, Restart and Rematch.
- Accessibility: Colour-blind mode and reduced-motion option.
- Localization: English, Hindi and Odia interface text.
- Audio: Optional SFX and Tanpura ambience.

Player Definition

Player 1 — Peacock / Side A

The Peacock side is the first player identity used by the game. It is represented through the player card and token set.

Player 2 — Haldi / Side B

The Haldi side is the second player identity. In Player vs AI mode, the computer controls this side.

Pieces

- Nine pieces are available to each player at the beginning of a match.
- Pieces have board positions, ownership and legal movement determined by the rules engine.

Rules and Gameplay

- Players alternate turns.
- Placement uses empty board points.
- A mill is a defined line containing three pieces of the same side.
- A successful mill creates a capture opportunity.
- Protected mill pieces are handled according to the implemented capture rule.
- Movement follows the board's legal adjacency connections.
- Flying becomes available at exactly three remaining pieces.
- The game checks both piece count and legal mobility for the end condition.

Player Rewards

- Mill formation creates an immediate capture opportunity.
- Legal-move indicators provide immediate visual feedback.
- Hint can suggest a useful move.
- Winning triggers celebratory confetti and result feedback.

User Interface (UI)

- Main 24-point board.
- Player status cards and piece information.
- Current turn and phase indicator.
- Legal destination markers.
- Undo and Hint buttons.
- Restart / Rematch controls.
- Settings modal with mode, AI difficulty, language, board variant, audio and accessibility controls.
- Rules/help screen explaining Placement, Movement and Flying.

Heads Up Display (HUD)

- Current turn.
- Current phase: Placement, Movement or Flying.
- Player-side information.
- Selected piece and valid destinations.
- Capture and win feedback.

Player View

The player sees the complete 2D board from a fixed top-down view. There is no scrolling character camera. Keeping the whole board visible supports strategic comparison of mills, threats and movement options.

Antagonistic Elements

The opponent is the main antagonistic element. In single-player mode the opponent is controlled by the local AI; in Pass & Play it is another human using the same device. There are no physical enemies, combat systems or environmental hazards.

Artificial Intelligence (AI)

- Easy: Shallow search with some random choices for a casual challenge.
- Medium: Depth-3 search with move ordering for a stronger tactical challenge.
- Master: Deeper search with iterative/deeper analysis and transposition caching.
- Evaluation: Considers material, mills, threats and mobility.
- Rule Safety: AI uses the same legal-move and rules logic as the human player.

Global Game Elements

- 24 playable points.
- Standard mill configuration plus optional diagonal variant.
- Three gameplay phases.
- Two player sides: Peacock and Haldi.
- Persistent language, audio and accessibility preferences.
- Rules, settings and result overlays.

Story and Theme

Dahdi is not a narrative adventure. Its identity comes from traditional board-game heritage and strategic play. The digital presentation aims to make the traditional rules accessible while preserving the focus on planning, alignment and tactical capture.

Level / Match Design

Dahdi is match-based rather than level-based. Each match contains Placement, Movement and Flying phases. The optional diagonal configuration changes board connectivity and possible mills without requiring separate story levels.

Audio & Sound F/X

- Web Audio API based interface and gameplay feedback.
- Soft UI tap and action sounds.
- Optional Indian Tanpura drone for ambient music.
- Separate SFX and music volume controls.
- Mute and preference persistence through browser localStorage.

Game Architecture

File / Module	Purpose
index.html	Main interface, rules/help text, settings and controls.
constants.js	Board points, players, phases, mills, translations and game constants.
rules.js	MillState / MillRules, board state, legal moves, mills, captures, protected pieces and win checks.
ai.js	Easy/Medium/Master search, evaluation, move ordering and transposition cache.
render.js	Board drawing, token rendering, legal move indicators and win confetti.
audio.js	Audio context, SFX, music and volume/mute management.
app.js	Application controller, state transitions, player interaction, settings, undo, hint and rematch.
CSS files	Main UI, board, tokens and design variables.

How to Play

- Start the game and choose Player vs AI or 2-Player Pass & Play.
- If using AI mode, choose Easy, Medium or Master.
- Place your nine pieces on empty board intersections.
- Try to align three pieces to form a mill.
- After a mill, capture an eligible opponent piece.
- After placement, move pieces along legal board connections.
- When you have exactly three pieces, use Flying to move to any empty point.
- Continue forming mills, capturing pieces and blocking the opponent.
- Win by reducing the opponent below three pieces or leaving no legal move.

Technical Document

- Technology: HTML5, CSS3 and JavaScript ES modules.
- Graphics: DOM/SVG/CSS based 2D board and tokens.
- Audio: Web Audio API.
- Input: Mouse/touch interaction and keyboard shortcuts.
- Storage: Browser localStorage.
- Network: No online multiplayer service required.
- External Engine: No game engine; browser-based implementation.

System Requirements

- Modern Chrome, Edge, Firefox or Safari browser.
- Desktop or mobile device capable of running modern JavaScript and CSS.
- Audio-capable device for sound features.
- Project files served/opened in a browser environment that permits JavaScript modules.

Programming Content

- State-driven game flow.
- Modular JavaScript game state and rules logic.
- 24-point adjacency and mill definitions.
- Legal move generation.
- Mill and protected-piece detection.
- Win-condition checking.
- Negamax / alpha-beta AI search.
- Move ordering and transposition caching.
- Event-driven UI controls.
- Web Audio API based sound management.

Resources

- HTML/CSS/JavaScript development environment.
- Modern web browser for testing.
- Dahdi project source files.
- Optional image/visual editing tools for future board assets.

Project Summary

Dahdi combines a compact traditional strategy rule set with a modular browser implementation. The final design separates rules, AI, rendering, audio and application control so that the project remains understandable, testable and extendable.

Final Design Intent

The completed design prioritizes rule clarity, strategic depth, accessibility and ease of demonstration. The result is intended to function as both a playable academic project and a digital presentation of a traditional Indian board-game style.

Game Pitch Proposal: Dahdi (Traditional Indian Mills Board Game)

Academic Game Development Project
Digital Traditional Strategy Game

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1. Cover Page

- Title of Project: Dahdi: Traditional Indian Mills Board Game
- Project Type: Digital Preservation & Casual Strategy Board Game
- Development Team: Student Game Development Project
- Lead Developer: Soumya Ranjan Samal
- Target Platforms: Modern Web Browser / HTML5 / CSS3 / JavaScript
- Date: September 2026

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2. Executive Summary

Dahdi brings a traditional Indian mills-style strategy game into a modern browser experience. The core game uses a 24-point board and nine pieces for each side. Players alternate between placing pieces, moving them and, when reduced to three pieces, flying them to any empty point. The central strategic objective is to create mills of three pieces and use them to capture the opponent.

The digital version adds a computer opponent with Easy, Medium and Master difficulty levels, a two-player Pass & Play mode, an optional diagonal board configuration, Hint and Undo functions, a Rules screen, audio and accessibility settings, and English, Hindi and Odia language options.

3. Game Description

a. Overview

Dahdi is a turn-based 2D strategy game for two sides. Each side has nine pieces and competes on a 24-point board. The game rewards planning rather than fast reflexes.

b. The Quest

The main quest is to outmaneuver the opponent by forming mills, capturing pieces, protecting your own formations and eventually reducing the opponent below three pieces or preventing all legal movement.

c. Main Character (The Player)

There is no narrative hero. The player's identity is represented through the Peacock side and the active player status card. The second side is represented by Haldi.

d. Sketch of Main Character / Token Layout

Insert a screenshot of the Peacock token and Player 1 status card from the game here.

e. Opponents (AI System)

The single-player opponent is the Haldi side controlled by the AI. Easy provides a casual challenge, Medium uses depth-3 search, and Master uses deeper search and caching.

f. Sketch of Opponent Architecture

Start Game → Determine Phase → Generate Legal Moves → Evaluate Candidate Moves → Search / Score Position → Select Move → Apply Move → Render Result.

g. Environment

The environment is a 2D traditional-inspired board with three concentric squares and 24 playable points. The interface uses a warm board-game presentation rather than a 3D world.

h. Sketch of Game Arena / World

Insert a screenshot of the complete 24-point Dahdi board here.

i. Menus

The interface includes game-mode selection, AI difficulty, diagonal-board option, language selection, audio controls, colour-blind mode, reduced motion, Rules, Settings, Undo, Hint, Restart and Rematch.

j. Controls & Ergonomics

Players select board points with mouse/touch interaction. Keyboard shortcuts include U for Undo, H for Hint, R for Rematch and Esc for Settings.

k. Sounds & Acoustic Identity

The game uses soft UI/gameplay feedback sounds and an optional Indian Tanpura drone. SFX and music volume can be controlled independently.

4. Strategic Experience

Dahdi is designed around readable decisions. Every turn asks the player to balance immediate opportunities against future board control. Forming a mill may provide a capture, but moving a piece can also open or close a future route. When a player reaches three pieces, the Flying phase changes the tactical possibilities because any empty point can become a destination.

5. Audience

- Students learning traditional games or game-development concepts.
- Casual players looking for a short strategy match.
- Families and friends using local Pass & Play.
- Players interested in Indian-inspired digital board games.

6. Accessibility & Localization

- English, Hindi and Odia interface text.
- Colour-blind mode.
- Reduced-motion option.
- Visible legal move indicators.
- Rules/help screen for first-time players.

7. Technical Pitch

- Frontend: HTML5 and CSS3.
- Logic: JavaScript ES modules.
- Rules: Centralized legal-move, mill, capture and win logic.
- AI: Easy, Medium and Master search with evaluation and caching.
- Rendering: 2D board and token rendering with legal-move feedback.
- Audio: Web Audio API.
- Persistence: Browser localStorage for selected preferences.

8. Why This Project Works as a College Game

The project demonstrates multiple software and game-development concepts in one manageable application: state management, rule modeling, search algorithms, UI interaction, accessibility, localization, audio programming and software modularity. The board-game scope keeps the project testable while still providing meaningful strategic depth.

9. Summary & Sales Pitch

Dahdi is a simple-to-access but strategically deep digital board game. Its identity comes from combining a traditional Indian mills-style rule set with a clean modern interface, a computer opponent, local two-player play and accessibility features. Because it runs in a browser, the game can be demonstrated easily in a classroom, lab or personal computer without requiring a large installation.

10. Demonstration Flow

- Open the game in a modern browser.
- Choose Player vs AI or 2-Player Pass & Play.
- Select AI difficulty if required.
- Place pieces and demonstrate a mill.
- Show the capture interaction.
- Move pieces during the Movement phase.
- Demonstrate Flying when a side has exactly three pieces.
- Open Settings and demonstrate language/accessibility options.
- Use Undo, Hint and Rematch.

11. Closing Statement

Dahdi is intended to show how a traditional strategic activity can be preserved through a lightweight digital implementation. The project balances authenticity of the core rules with practical modern features that make the game easy to demonstrate, understand and replay.

Project Charter: Dahdi (Traditional Indian Mills Board Game)

Academic Game Development Project
Browser-Based Strategy Game

Charter Field	Project Specification
Project Name	Dahdi: Traditional Indian Mills Board Game
Project Description	Dahdi is a browser-based turn-based strategy board game based on the Indian mills/alignment game family. It uses a 24-point board, nine pieces per side, mill formation, captures, movement and a flying endgame. It supports Player vs AI and 2-Player Pass & Play.
Project Sponsor	Academic / College Game Development Project
Department / Context	Game Development / Academic Project
Project Team Member(s)	Soumya Ranjan Samal
Platform	Web Browser / HTML5 / CSS3 / JavaScript

Project Objectives

- Create a playable and understandable Dahdi game.
- Implement placement, movement, flying, mill, capture and win rules consistently.
- Provide Easy, Medium and Master AI opponents.
- Provide local two-player Pass & Play.
- Design a clear UI with turn and phase feedback.
- Add sound, localization and accessibility features.
- Test the complete game and prepare academic documentation.

Project Milestones

- M0 – Concept and Rules: Define 24 positions, nine pieces, mills, captures, phases and win conditions.
- M1 – Board and UI: Implement board, tokens, player cards, phase indicator, buttons and modals.
- M2 – Rules Engine: Implement legal moves, captures, protected pieces and end conditions.
- M3 – AI: Implement Easy, Medium and Master search and evaluation.
- M4 – Audio, Settings and Accessibility: Implement Web Audio, language preferences, colour-blind mode and reduced motion.
- M5 – QA: Test each phase, captures, AI moves, controls, settings and result states.
- M6 – Documentation: Complete GDD, pitch, WBS, journal, charter and modification sheet.

Assumptions and Constraints

- No online multiplayer backend is included.
- The game is designed around a 24-point board.
- Each player has nine pieces.
- The AI is local and does not require an external service.
- Browser compatibility depends on support for modern JavaScript modules and Web Audio.
- Future visual expansion can be added without changing the core rules module.

Project Completion Date

Academic submission period — September/October 2026.

Project Authorization & Sign-Off

Project Sponsor	Academic / College Project
Student Developer	Soumya Ranjan Samal
Status	Active / Academic Submission
Review Date	September 2026
Approval	To be completed by faculty / project supervisor

Authorization Note

This charter defines the scope and intended academic deliverables of the Dahdi project. Changes to core rules, platform or major milestones should be documented before final submission.

Work Breakdown Structure (WBS): Dahdi Game

Project Name: Dahdi — Traditional Indian Mills Board Game
Institution: Academic / College Game Development Project
Author / Lead Developer: Soumya Ranjan Samal
Collection: Traditional / Indigenous Strategy Game Project

Level	WBS Code	Element Name (Development Steps)	Definition (Clearly define work to be done)
1	1.0	**Project Startup & Framework Initialization**	Initial project scoping, technical requirements, source structure and browser development setup.
2	1.1	Game Concept & Rules Formalization	Define Dahdi objectives, nine-piece setup, 24 board points, mills, captures, movement, flying and win conditions.
2	1.2	Architecture Design & Module Separation	Separate application control, rules, AI, rendering, audio, constants and styles.
2	1.3	Source Organization & Development Setup	Maintain JavaScript modules and CSS files in organized project directories and test the game in a modern browser.
1	2.0	**Board & Core Game Mechanics**	Implement the playable 24-point board and the complete match state.
2	2.1	24-Point Board Configuration	Define the three concentric-square board positions and point coordinates.
2	2.2	Player Piece State	Create nine pieces per side and track pieces in hand and on the board.
2	2.3	Placement Phase	Allow players to place one available piece on an empty point during Phase 1.
2	2.4	Movement Phase	Generate and apply legal adjacent moves during Phase 2.
2	2.5	Flying Phase	Allow a player with exactly three pieces to move to any empty point during Phase 3.
2	2.6	Mill & Capture Rules	Detect three-in-a-row mills, identify removable opponent pieces and protect pieces in mills when required.
2	2.7	Win & Draw Conditions	Detect opponent reduction to two pieces and situations with no legal movement.

Work Breakdown Structure (WBS): Dahdi Game

Level	WBS Code	Element Name (Development Steps)	Definition (Clearly define work to be done)
1	3.0	**Artificial Intelligence (AI) Opponent**	Develop the local computer opponent and difficulty settings.
2	3.1	Easy AI	Use shallow search and some random choices for a casual challenge.
2	3.2	Medium AI	Use depth-3 search and move ordering for a balanced challenge.
2	3.3	Master AI	Use deeper search, iterative/deeper analysis and transposition caching.
2	3.4	Position Evaluation	Evaluate material, mills, threats and mobility.
2	3.5	Hint System	Use the AI decision system to provide a useful player move hint.
1	4.0	**Art, Visuals & Game Feedback**	Create the visual board, tokens, indicators and celebratory effects.
2	4.1	Board Rendering	Render the 24-point board and selectable positions.
2	4.2	Player Tokens	Display Peacock and Haldi pieces and player status.
2	4.3	Legal Move Indicators	Highlight selectable pieces and valid destination points.
2	4.4	Victory Feedback	Trigger win confetti and result feedback.
1	5.0	**Audio & Settings**	Implement game audio, music and player preferences.

Work Breakdown Structure (WBS): Dahdi Game

Level	WBS Code	Element Name (Development Steps)	Definition (Clearly define work to be done)
2	5.1	Web Audio Engine	Initialize audio context and separate SFX/music gain controls.
2	5.2	Sound Effects	Provide UI and gameplay feedback sounds.
2	5.3	Tanpura Music	Provide optional ambient Indian Tanpura drone.
2	5.4	Preference Storage	Store mute, volume, language, colour-blind and reduced-motion preferences in local storage.
1	6.0	**User Interface & User Experience**	Build menus, settings, help, HUD and accessibility presentation.
2	6.1	Game Mode Selection	Provide Player vs AI and 2-Player Pass & Play.
2	6.2	AI Difficulty Selection	Provide Easy, Medium and Master.
2	6.3	Board Variant	Provide the optional diagonal configuration.
2	6.4	Language Selection	Provide English, Hindi and Odia.
2	6.5	Rules / Help Screen	Explain Placement, Movement and Flying.
2	6.6	Undo / Hint / Rematch	Provide utility controls for player assistance and replay.
1	7.0	**Quality Assurance & Final Submission**	Verify gameplay and prepare final academic documentation.
2	7.1	Rules Testing	Test placement, movement, flying, mills, captures and win conditions.

Work Breakdown Structure (WBS): Dahdi Game

Level	WBS Code	Element Name (Development Steps)	Definition (Clearly define work to be done)
2	7.2	AI Testing	Test all AI levels and ensure only legal moves are selected.
2	7.3	UI Testing	Test settings, language, audio, accessibility, undo, hint and result states.
2	7.4	Browser Testing	Check rendering and interaction in modern browsers.
2	7.5	Documentation	Prepare GDD, Pitch Proposal, Project Charter, WBS, Journal and Modification Task Sheet.

Game Modification Task Sheet: Dahdi

Institution: Academic / College Game Development Project
Project: Dahdi – Traditional Indian Mills Board Game
Review Date: September 2026
Evaluator: Faculty / Project Supervisor

Game Reviewed:

Dahdi (Traditional Indian Mills / Alignment Strategy Game)

Game Platform:

Web Browser / HTML5 / CSS3 / JavaScript

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1. Provide a detailed overview of the object of the game.

The fundamental object of Dahdi is to defeat the opposing player through strategic placement and movement of pieces. Each player has nine pieces. Players form mills of three aligned pieces and use successful mills to capture eligible opposing pieces. After placement, the game moves into a movement phase and, when a player has exactly three pieces, a flying phase becomes available. The game ends when the opponent has fewer than three pieces or no legal move.

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2. List 3 or more things you like about the game. Why?

- Clear strategic objective: Mills and captures give every turn a meaningful purpose.
- Simple interaction: The board has a limited number of points, making the controls easy to understand.
- Multiple AI levels: Easy, Medium and Master provide different levels of challenge.
- Local multiplayer: Pass & Play allows two people to use the same device.

3. List things you do not like about the game. Why?

- New-player learning curve: Mills, protected pieces and Flying may be unfamiliar to first-time players.
- No online multiplayer: The current project does not support remote matches.
- Limited personalization: The board has only the implemented visual configuration and optional diagonal variant.
- Limited long-term statistics: The current version does not maintain a full match-history dashboard.

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4. What changes would you make to improve the game?

Modification	**Problem Solved**	**Implementation Detail**
Tutorial / Guided First Match	Reduces confusion for new players.	Add step-by-step prompts for placement, mill formation, capture, movement and Flying.
Online Multiplayer	Adds remote competition.	Add a future server-based multiplayer layer while keeping the existing rules engine reusable.
Enhanced Board Animation	Improves visual feedback.	Add smoother piece movement, capture transitions, mill highlights and phase transitions.

Modification Plan — Continued

Modification	Problem Solved	Implementation Detail
Difficulty Customization	Gives more control over AI challenge.	Add selectable search depth or additional AI behaviour profiles beyond Easy/Medium/Master.
Statistics & Match History	Improves replay value.	Store wins, losses, moves, average match length and selected difficulty in local browser storage.
Theme / Board Skins	Adds personalization.	Add optional traditional visual themes while keeping the current default design.
Guided Hint Explanations	Makes the Hint feature more educational.	Show why a suggested move creates a threat, blocks a mill or improves mobility.

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5. Priority for Future Versions

- Guided tutorial and clearer first-match explanations.
- Improved visual transitions and board animation.
- Optional statistics and match history.
- Additional AI configuration and difficulty controls.

6. Final Modification Assessment

The current Dahdi implementation already contains the essential board-game loop, multiple AI levels, local multiplayer, accessibility controls, localization, audio, undo, hint and rematch functionality. The suggested modifications are therefore focused on improving onboarding, replay value, personalization and future connectivity rather than replacing the core game.

7. Recommended Development Sequence

- First: add a guided tutorial.
- Second: improve board animation and capture feedback.
- Third: add statistics and match history.
- Fourth: add optional visual themes.
- Fifth: evaluate online multiplayer as a separate future module.

Conclusion

The modification plan preserves the existing rules and modular architecture of Dahdi. Each proposed improvement can be added without changing the central board-state and rules engine, allowing the academic version to remain stable while providing a clear path for future development.

Game Developer’s Journal: Dahdi

Project: Dahdi – Traditional Indian Mills Board Game
Developer / Designer: Soumya Ranjan Samal
Development Period: 2026
Academic Project: College Game Development Project

Developer's Journal Manifesto

"A good game requires a sound strategy for success. The Game Developer's Journal serves as written evidence of strategic thinking, creative decisions, technical work, problem solving and refinement completed during development."

This journal records the major development stages represented by the final Dahdi project, including game selection, rule modeling, AI development, rendering, audio, accessibility and testing.

Journal Entry 1: Project Genesis & Game Selection

Phase: Concept & Research

The project began with the decision to create a traditional Indian-inspired board-game experience in a digital format. Dahdi was selected because the mills-style rule system has a clear board, a small number of pieces and meaningful strategic interaction. The goal was to keep the rules understandable while adding modern digital usability.

The initial design target was a browser game that could be demonstrated without a large installation. This led to a modular HTML/CSS/JavaScript architecture.

Journal Entry 2: Board Structure & Rule Modeling

Phase: Game Rules

The board was modeled as 24 playable points arranged across three concentric squares. The rules structure was separated from the visual interface so that placement, movement, mills, captures and win conditions could be tested independently. Nine pieces per player were defined as the standard setup.

Journal Entry 3: Placement, Movement & Flying

Phase: Core Gameplay

The game was divided into three phases. Placement allows players to put their pieces onto empty points. Movement allows legal adjacent movement after placement. Flying becomes available when a player has exactly three pieces, allowing movement to any empty point. This state-driven design makes the match flow predictable and easy to display in the UI.

Journal Entry 4: Mill Detection & Capturing

Phase: Rules Engine

The rules engine checks the mills passing through each board point. When a player forms a mill, the game identifies valid opponent pieces for removal. Protected pieces inside mills are handled according to the implemented capture rule, while a fallback is provided when all opponent pieces are protected.

Journal Entry 5: Artificial Intelligence

Phase: AI Development

A local AI opponent was added with three difficulty levels. Easy uses shallow search and some random choices. Medium uses depth-3 search and move ordering. Master uses deeper search with iterative/deeper analysis and a transposition cache. The evaluation function considers piece count, mills, threats and mobility.

Journal Entry 6: Rendering & User Feedback

Phase: Visual Development

The board renderer was separated from the rules engine. The interface highlights selectable pieces and legal destinations, shows the current phase and turn, and provides visual feedback when a match is won. Confetti was added as a lightweight celebratory effect.

Journal Entry 7: Audio System

Phase: Audio Development

The audio module was implemented using browser audio capabilities. It supports interface and gameplay sound effects and an optional Tanpura drone. Separate sound and music volumes allow the player to adjust the experience without changing the game rules.

Journal Entry 8: Settings, Localization & Accessibility

Phase: UX Development

The settings system was expanded with Player vs AI, Pass & Play, AI difficulty, diagonal board option, language selection, Tanpura music, SFX controls, colour-blind mode and reduced-motion mode. English, Hindi and Odia interface strings were included.

Journal Entry 9: Undo, Hint & Replay

Phase: Player Support

Undo allows the player to reverse an earlier action, while Hint uses the AI decision system to suggest a useful move. Rematch provides a quick route back into a new match after a result. These utilities improve experimentation and make the game easier to demonstrate.

Journal Entry 10: Testing & Final Refinement

Phase: Quality Assurance

Testing focused on placement, movement, flying, mill detection, protected captures, AI legality, UI controls, language settings, audio, accessibility and result states. The final refinement pass emphasized clear feedback and keeping the rules engine authoritative.

Current Project Reflection

Dahdi developed from a simple traditional board-game concept into a modular browser application with a rules engine, AI, rendering, audio and accessibility systems. The strongest technical lesson was the value of separating game rules from presentation. This separation made it easier to reason about legal moves, mill formation, captures and AI decisions.

The project also demonstrated that a small board can still create substantial strategic depth. The same 24 positions support placement tactics, blocking, mill threats, mobility decisions and a different endgame once Flying becomes available.

Final Development Note

The completed project provides a stable academic foundation. Future work can focus on guided tutorials, stronger visual animation, statistics, board themes and optional online multiplayer without replacing the existing core rules architecture.

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Project: Dahdi – Traditional Indian Mills Board Game
Year: 2026